

Recent Developments in Data Science: Comparing Linear, Ridge and Lasso Regressions Techniques Using Wine Data

Mayooran Thevaraja¹

Lecturer in Engineering Mathematics,
Department of Interdisciplinary studies,
Room 38, Faculty of Engineering, University of Jaffna,
Ariviyal Nagar, Killinochchi, Sri Lanka.
mayooran@eng.jfn.ac.lk

Azizur Rahman²

Senior Lecturer in Mathematics and Statistics
Charles Sturt University
Room 310 | B001, Boorooma Street
Wagga Wagga, NSW 2678, Australia
azrahman@csu.edu.au

Mathew Gabirial³

Department of Mathematics and Statistics,
Minnesota State University, Mankato, USA.
mathew.gabriel@mnsu.edu

ABSTRACT

Big data is the reality of the 21st century. However, big data modeling and prediction require advanced level analytics which encompasses both the computing-intensive and statistics-oriented analysis tools in data science. Regression analysis is the statistical method for predictive modeling, and it is one of the most commonly used methods in many scientific fields such as engineering, the physical and chemical sciences, economics, management, life and biological sciences, and the social sciences, sociology, geology, etc. Satisfying the assumptions such as collinearity between variables ought to be a significant issue in data science. Advanced level tools such as Lasso and Ridge regression methods are designed to overcome such problem. In this study we discussed about comparing linear regression with the Ridge and Lasso. The Vinho Verde white wine test data from the Minho (northwest) region of Portugal is used to analyze advantages of each of the three regression analysis methods. All the required calculations and graphical displays are performed using the R software for statistical computing.

Keywords

Standard regression, Ridge, Lasso, White wine data, Validation of estimates.

1. INTRODUCTION

In data science, the analyst might be interested in the estimated values for the objects in the training set, predicting responses for new objects. However, identifying which input variables are most important in making good predictions, is challenging due to complex relations between these variables (Rahman, 2008). For example, in wine industry physicochemical laboratory tests are routinely used to characterize quality of wine such as its density, alcohol or pH values. Also sensory tests are required which mainly

depend on experts people within the industry. Classification of wine is a challenging task though. The links between the physicochemical and sensory analysis are fairly complex and require advance level modelling to understand them better (Legin et al. 2003).

A big dataset can preserve useful facts and figures which can be utilized to effective decision making process (Rahman, 2017). For instance, data mining techniques including the linear regression focus on extracting advance level knowledge by modelling raw data with continuous variables (Turban et al. 2007; Rahman and Harding, 2016). The performance of any data mining tools also very much depends on a right set variables and robust model selection (Das et al. 2018; Chowdhury et al. 2018; Rahman et al. 2018). Typically, a simple model such as linear regression may fail in mapping the underlying concept due to collinearity and too complex ones such as generalized regression tend to over fit the data due to lack of functional link and collinearity as well (Hastie et al. 2001; Guyon and Elisseeff 2003). Often the traditional regression model can show higher correlation coefficients, but can suffer from overfitting and its lower prediction accuracy (Alexandre-Tudo et al. 2015).

One of the ways of avoiding overfitting is using cross validation that helps in estimating the error over test data set, and in deciding what parameters work best for your model. For examples, regularized regression techniques such as Lasso and Ridge methods are designed to minimize the consequences of collinearity between variables by avoiding overfitting and also increasing model interpretability with an increased level of accuracy. Regularization process imposes an upper threshold on the values taken by the coefficients that constrains or shrinks the coefficient estimates towards zero especially for the variables which influence the overfitting.

Some studies related to the Lasso technique have primarily focused on available bench-mark data sets. In particular, Sardy and Bruce (1998) use the method to detecting the incoming radar signatures. Zheng et al. (2002) works basis pursuit to epidemiological studies by applying logistic regression based Lasso for identifying features associated with program crashes. Several extensions of the Lasso technique in the literature tweak the manner in which the coefficients are constrained. One extension is the group Lasso, which looks at groups of variables jointly (Meier et al. 2008). The group Lasso and some of its variants have been widely used in genetic association and genome-wide association studies. This method allows groups or clusters of inputs to be in or out of the model together (Liu and Zhang, 2009). However, the Lasso method and its derivatives are not always consistent for variable selection (Meinshausen and Yu, 2009).

More recently, Mayoaran (2018) studies simple linear regression and ridge regression parameter estimations via the gradient-based optimization algorithm for the ridge regression technique by using R. Ridge approach (Hoerl et al. 1970) is ideal if there are several predictors come from a normal population and all of them have non-zero coefficients (Friedman et al. 2010). In particular, it performs well with many predictors each having small effect and prevents coefficients of linear regression models with many correlated variables from being poorly determined and exhibiting high variance (Ogut et al. 2012). Ridge regression shrinks the coefficients of correlated predictors equally towards zero. So, for example, given k identical predictors, each would get identical coefficients equal to $1/k$ the size that any one predictor would get if fit singly (Friedman et al. 2010). This technique thus does not force coefficients to vanish and hence cannot select a model with only the most relevant and predictive subset of predictors.

Thus this paper aims to compare the classical linear regression method with the advanced regularized regressions such as Ridge and Lasso techniques.

2. METHODOLOGY

2.1 Linear regression model

In the typical univariate linear regression scenario, the input is an instances of a process described by $p + 1$ variables, and we seek to find a linear combination n of a selected p variables (the features) that best predicts the remaining variable (the target). We typically define the design matrix X as a matrix having n rows and p columns, representing the p variables for n instances. Similarly, we define the target vector y as a column vector of length n containing the corresponding values of the target variable. The problem can then be formulated as finding a “good” value for the length p coefficient vector β : In general, the response y may be related to k regressors or predictor variables. The model is given by,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_n x_n + \epsilon \cdots \cdots \cdots (1)$$

Multiple linear regression models are often used as approximating functions. We can use ordinary least square method for this parameter estimation which is usual way so in this section we are going to discuss result in matrix format equation. It is more convenient to deal with multiple regression models if they are expressed in matrix notation (see, e.g. Rahman, 2008). This allows a very compact display of the model, data, and results. In matrix notation, the model given by is

$$y = X\beta + \epsilon \cdots \cdots \cdots (2)$$

By the least square method, we can get that solutions

$$\hat{\beta} = (X^T X)^{-1} X^T y \cdots \cdots \cdots (3)$$

2.2 Ridge technique

The most popular form of regularized regression is Ridge Regression, which places a constraint on the sum of squares of the coefficients weights through a constraint on the p coefficients. Ridge regression can be formulated as follows,

$$\hat{\beta}_{ridge} = \arg \min_{\beta \in R^p} \sum_{i=1}^n (y_i - x_i^T \beta)^2 \cdots \cdots \cdots (4)$$

with subject to $\sum \beta_j^2 \leq t$ for $t \geq 0$.

The feasible set for this minimization problem is therefore constrained to be $S(t) = \{\beta \in R^p : \|\beta\| \leq t\}$ where β does not include the intercept β_0 . The ridge estimator are not equivariant under a rescaling of the x_j 's, because of the L2-penalty. This difficulty is circumvented by centering the predictors, and can be formulated using a Lagrange multiplier as

$$\hat{\beta}_{ridge} = \arg \min_{\beta \in R^p} \left\{ \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\} \cdots \cdots \cdots (5)$$

The optimization problem in (5) can be seen to be a convex Minimization in β . Note that the Residual Sum of Squares i.e. RSS is convex in β , provided that $X^T X$ is positive definite, which follows from the fact that X is assumed to be full-rank. Moreover, one can easily show that the sum of square β_j 's is also convex. The RSS for ridge regression can be expressed as,

$$RSS(\beta; \lambda) = (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta \cdots \cdots \cdots (6)$$

One can minimize this criterion using straightforward applications of matrix calculus, as was conducted for the classical OLS criterion for multiple regression. That is, taking the first derivative and setting to zero, we obtain

$$\frac{\partial}{\partial \beta} RSS(\beta; \lambda) = 2(X^T X)\hat{\beta} - 2X^T y + 2\lambda \hat{\beta} = 0 \cdots \cdots \cdots (7)$$

This expression can be further simplified as follows,

$$(X^T X + \lambda I)\hat{\beta} = X^T y \cdots \cdots \cdots (8)$$

and therefore, the ridge estimators are

$$\hat{\beta}_{ridge} = (X^T X + \lambda I)^{-1} X^T y \cdots \cdots \cdots (9)$$

Since we are adding a positive constant to the diagonal of $X^T X$, we are, in general, producing an invertible matrix, $X^T X + \lambda I$, even if $X^T X$ is singular. It can be easily shown that the ridge regression estimator is related to the classical OLS estimator. If $X^T X$ is non-singular, the OLS estimation relationship can be verified as well. Also when X is composed of orthonormal variables, such that $X^T X = I_p$, it then follows that

$$\hat{\beta}_{ridge} = \frac{1}{1 + \lambda} \hat{\beta}_{OLS} \cdots \cdots \cdots (10)$$

This shows that the ridge estimator is simply a down weighted version of the OLS estimator.

2.3 Lasso technique

The lasso regression technique tries to produce a sparse solution, in the sense that several of the slope parameters will be set to zero. The Lasso is also formulated with respect to the center matrix, X . Moreover, the L1-penalty is solely applied to the slope coefficients, and thus the intercept, β_0 , is excluded from the penalty term. Hence the Lasso can be expressed as a constrained minimization problem,

$$\hat{\beta}_{lasso} = \arg \min_{\beta \in R^p} \sum_{i=1}^n (Y_i - X_i^T \beta)^2 \dots \dots \dots (11)$$

subject to

$$\sum_{j=1}^p |\beta_j| \leq t$$

for $t \geq 0$; which can again be re-formulated using the Lagrangian for the L1- penalty, as follows,

$$\hat{\beta}_{lasso} = \arg \min_{\beta \in R^p} \left\{ \sum_{i=1}^n (Y_i - X_i^T \beta)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\} \dots \dots \dots (12)$$

Where $\lambda \geq 0$ and, as before, there exists a one-to-one correspondence between t and λ . Contrary to ridge regression, the Lasso does not admit a closed-form solution. The L1-penalty makes the solution non-linear in the y_i 's. The above constrained minimization is a quadratic programming problem, whose solution can be efficiently approximated.

2.4 Cross-validation of Ridge and Lasso

Cross-validation is an appropriate method to find a “most suitable” estimate for the λ parameter with a given dataset. Typically, λ values that are too small can lead to over fitting when the model would tend to describe the noise in the data. On the other side, large λ values would lead to under fitting when the procedure cannot capture the underlying relationship. To perform cross validation, first the initial data set is randomly divided into K blocks of equal length. One of the blocks is assigned the role of the test set while the remaining $K - 1$ blocks together constitute the train set. In practice the number of blocks K is usually selected to be 5 or 10. Next we choose a grid of values $\lambda = \lambda_s$ and calculate the regression coefficients for each λ_s value. Given these regression coefficients, we then compute the residual sum of squares.

$$RSS_{\lambda_s, k} = \sum_{i=1}^n \left(y_i - \sum_{j=1}^{k-1} \hat{\beta}(k, \lambda_s) x_{ij} \right)^2 \dots \dots \dots (13)$$

where $k = 1, \dots, K$ is the index of the block selected as the test set. One can obtain the average of these RSS values over all blocks.

$$MSE_{\lambda_s} = \frac{1}{K} \sum_{k=1}^K RSS_{\lambda_s, k} \dots \dots \dots (14)$$

λ is then set equal to λ_s that gives the minimum MSE_{λ_s} . Second popular approach utilizes the “one-standard-error rule”. For each MSE_{λ_s} the standard error of the mean is calculated. Then the largest λ_s is selected for which MSE_{λ_s} is within one standard error of the

minimum MSE value. This way we get a “more regularized” model while increasing the MSE by not more than one standard error.

3. EMPIRICAL RESULTS

3.1 Data

In this study, we will be using observations of wine data which is collected from UCI Machine Learning Repository. The Wine Quality data set describes red and white variants of the “Vinho Verde” wine from the Minho province in the north of Portugal. In this work, we only analyze the white wine data by considering the 11 physiological predictors which encompass the physical and chemical characteristics of the “Vinho Verde”. The wine quality score, an integer value between 0 and 10 that we consider as the response variable. We build regression models that describe how the wine quality score depends on the physiological characteristics of the wine using OLS, Ridge and Lasso. We use R software for all calculations, namely its `glmnet` package.

3.2 Linear regression

To be able to measure quality of the regression model, we randomly split the whole sample into two subsets: the train set and the test set. Later we will validate our results on a test set. First, we are going to fit the linear fitted line equation for train set. According to table 1, β_3, β_5 and β_7 are not significant variables.

Table 1. Coefficient and VFI estimates for the white wine data for OLS regression techniques-train set.

Variable	Coef.	t-test	p-value	VFI
constant	$\hat{\beta}_0 = 195.2$	5.97	0.000	
fixed.acidity	$\hat{\beta}_1 = 0.109$	3.28	0.001	3.30
volatile.acidity	$\hat{\beta}_2 = -1.964$	-11.82	0.000	1.14
citric.acid	$\hat{\beta}_3 = -0.031$	-0.23	0.819	1.15
residual.sugar	$\hat{\beta}_4 = 0.100$	8.15	0.000	15.4
chlorides	$\hat{\beta}_5 = 0.053$	0.06	0.949	1.26
freesulfur.dioxide	$\hat{\beta}_6 = 0.004$	3.01	0.003	1.82
totalsulfur.dioxide	$\hat{\beta}_7 = 0.000$	0.04	0.966	2.38
density	$\hat{\beta}_8 = -195.8$	-5.91	0.000	39.3
pH	$\hat{\beta}_9 = 0.768$	4.75	0.000	2.52
sulphates	$\hat{\beta}_{10} = 0.583$	3.85	0.000	1.18
alcohol	$\hat{\beta}_{11} = 0.156$	3.73	0.000	11.2

To check the train data for multicollinearity, we have to calculate the variance inflation factors VIF_j . The VIF_j values are also provided in the table 1. Some of these values are rather high, like $VIF_{r.sug}$, VIF_{dens} and VIF_{alc} which indicate multicollinearity exist in this model.

3.3 Ridge and Lasso models

Now we are going to fit the Ridge regression model for wine data (train set). We want to choose λ large enough to provide stable coefficients, but not unnecessarily large ones, as this introduces

additional bias and increases the residual mean square. This study uses popular cross validation method for estimated λ . The figure 1 depict the mean squared prediction error against $\log(\lambda)$. The gray bars at each point show MSE plus and minus one standard error (Interval). One of the vertical dashed lines shows the location of the minimum of MSE. The second dashed line shows the point selected by the one-standard-error rule. Using the cross-validation results, we now get proper values for $\log(\lambda)$. In this case R programming gives that exact proper λ values `ridge.bestlam` (i.e. $\text{Ridge}_{\lambda_{best}}$) = 0.04364007 and `ridge.lamlse` (i.e. $\text{Ridge}_{\lambda_{lse}}$) = 0.2328939. It is apparent that the mean squared prediction errors are fairly large for a high value of $\log(\lambda)$.

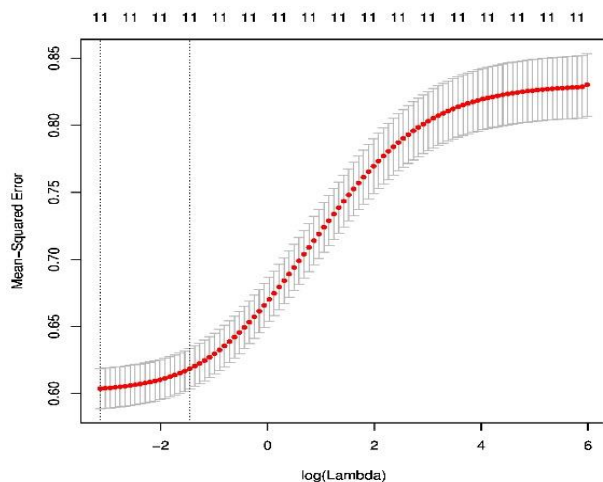


Figure 1. Cross-validated estimate of the mean square error for Ridge

Next, we fit the Lasso regression model for white wine data (train set). The figure 2 depict the mean squared prediction error against $\log(\lambda)$. Using the cross-validation results, we now get proper values for Lasso model as λ values `lasso.bestlam` (i.e. $\text{Lasso}_{\lambda_{best}}$) = 0.0007805678 and `lasso.lamlse` (i.e. $\text{Lasso}_{\lambda_{lse}}$) = 0.02938788. Graph shows that the mean squared prediction errors are dramatically increasing with the $\log(\lambda)$ value of around -4.5 or more.

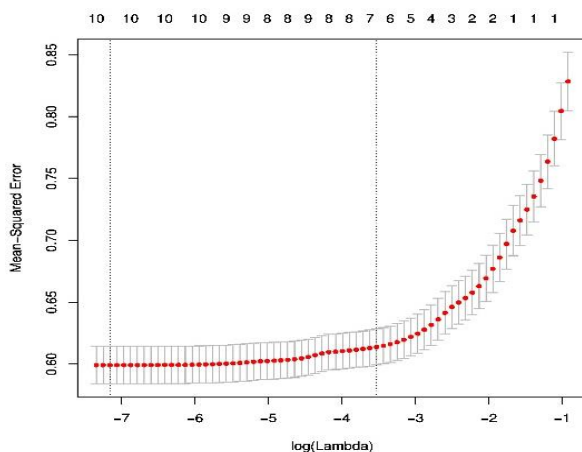


Figure 2. Cross-validated estimate of the mean square error for Lasso

3.4 Comparing results

In this section, we are going to compare the all three models by using RSS. First we estimated λ values based on train data sets by using two cross validation. Then we apply our estimated λ values in train data for estimating coefficients from Ridge and Lasso models. Results are presented in table 2. Data demonstrate that when λ value increases to positive direction, estimated coefficient values – especially for the Lasso model tends to converge to zero.

Table 2. Coefficient estimates for the white wine data for OLS and regularized regression techniques-train set.

Coef	OLS	$\text{Ridge}_{\lambda_{best}}$	$\text{Ridge}_{\lambda_{lse}}$	$\text{Lasso}_{\lambda_{best}}$	$\text{Lasso}_{\lambda_{lse}}$
$\hat{\beta}_0$	195.2	73.151	42.199	177.771	2.773
$\hat{\beta}_1$	0.109	0.006	-0.028	0.009	-0.029
$\hat{\beta}_2$	-1.964	-1.879	-1.472	-1.965	-1.766
$\hat{\beta}_3$	-0.031	-0.028	0.035	-0.025	0.000
$\hat{\beta}_4$	0.100	0.051	0.026	0.094	0.019
$\hat{\beta}_5$	0.053	-0.974	-2.227	-0.025	-0.314
$\hat{\beta}_6$	0.004	0.004	0.004	0.004	0.003
$\hat{\beta}_7$	0.000	-0.001	-0.001	0.000	0.000
$\hat{\beta}_8$	-195.8	-0.716	-39.135	-178.08	0.000
$\hat{\beta}_9$	0.768	0.321	0.183	0.698	0.000
$\hat{\beta}_{10}$	0.583	0.370	0.246	0.548	0.000
$\hat{\beta}_{11}$	0.156	0.271	0.226	0.175	0.344

Then we apply that estimated λ values in test data and also estimate coefficients and calculated RSS for each models. As can be seen, both Ridge and Lasso produce a more regularized model when moving from λ_{best} to λ_{lse} in the sense that the coefficient estimates get more “shrunk” towards zero. In case of Lasso, the number of nonzero coefficients changed. The coefficient $\hat{\beta}_7$ values remains zero at λ_{best} and λ_{lse} while coefficients $\hat{\beta}_3, \hat{\beta}_8, \hat{\beta}_9$ and $\hat{\beta}_{10}$ becomes equal to zero. Then if we apply our estimated λ values in test data set, we can get new estimated coefficient values which are given in a table A1 in the appendix. Data show differences in the estimated coefficients of the parameters. Moreover, to compare the performance of all models, first calculate the RSS on the train set and then move to the test set. The residual sum of squares (RSS) values for all the methods are given in table 3.

Table 3. RSS values for the train and the test sets by different regression techniques.

RSS	$\text{RSS}_{\text{train}}$	RSS_{test}
OLS	1449.813	1316.359
$\text{Ridge}_{\lambda=0.04364}$	1460.769	1311.863
$\text{Ridge}_{\lambda=0.23289}$	1500.089	1334.390
$\text{Lasso}_{\lambda=0.000781}$	1450.012	1314.046
$\text{Lasso}_{\lambda=0.02939}$	1489.077	1339.885

Here the smallest RSS value on the train set is predictably achieved by the OLS regression since unlike Ridge and Lasso the OLS does not impose penalties on the coefficients. RSS for regularized models is higher when λ is selected using the one-standard error rule. The errors on the test set are ordered differently the minimum is achieved by Ridge at 0.04364 (i.e. $\text{Ridge}_{\lambda_{\text{best}}}$), the second is Lasso at $\lambda = 0.000781$ (i.e. $\text{Lasso}_{\lambda_{\text{best}}}$) while OLS takes only the third position. In this case, both the regularized models perform better than OLS model.

4. ADVANTAGES AND LIMITATION

This section provides advantages and limitation of this research. A standard linear regression model tends to have some variance in it in many big data studies. Typically, this model won't generalize well for a data set different than its training data. Regularization, significantly reduces the variance of the model, without substantial increase in its bias. So the parameter λ , used in the Ridge and Lasso regularization techniques, controls the impact on bias and variance. Additionally, the primary difference between Ridge regression and the Lasso is that the first one constrains the L2 norm (sum of squares) of the coefficients, whereas the Lasso constrains the L1 norm (sum of absolute values) of the coefficients. Ordinary least-squares regression is a special case for both methods when λ equals 0. Both methods have the effect of forcing some of the coefficients toward zero, but at different rates.

Moreover, the number of variables that are forced to 0 depends on the value of λ (larger λ implies more coefficients forced to 0), and thresholding on λ allows the selection of variables that have retained nonzero coefficients at that λ threshold. However, a limitation is that after certain value of λ , the model starts losing important properties by losing related variables, giving rise to bias in the model and thus under fitting the model. Therefore, the value of λ should be carefully selected for a robust model.

5. CONCLUSIONS

The Ridge regression and the LASSO tend to “shrink” to zero coefficient estimates in the sense that they reduce the norm of the estimate vector as λ increases. The Ridge regression does not produce zero estimates even for large values of λ . The Lasso performs variable selection i.e. some of the coefficient estimates become exactly equal to zero, which makes the regression model easier to interpret. The RSS value calculated on the train set for OLS is less than that for Ridge and Lasso. When λ is selected appropriately, the RSS value on the test set for OLS is greater than the RSS for Ridge and Lasso. Calculating the RSS on the test data set provides a good way to assess the regression model in contrast to using a single data set. We can apply our result for several variety of data sets. In every considered test data, if we get Lasso and Ridge models have less RSS values, then we can conclude that regularized models are best fitting models in regression analysis when we found noise exists in the usual models.

6. ACKNOWLEDGMENTS

We would like to thank Iresha Premaratna at the Minnesota State University for the assistance during developing this manuscript.

Thanks are also due to the anonymous referees for their stimulus and valuable comments which were used to improve the paper.

7. REFERENCES

- [1] Aleixandre-Tudo, J. L., Alvarez, I., Garcia, M. J., Lizama, V., and Aleixandre, L. J. 2015. Application of multivariate regression methods to predict sensory quality of red wines. *Czech J. Food Sci.* 33: 217–227.
- [2] Chowdhury, M., Rahman, A., and Islam, R. 2018. Malware analysis and detection using data mining and machine learning classification. In *Advances in Intelligent Systems and Computing*. 580: 266–274.
- [3] Cortez, P., Cerdeira, A., Almeida, F., Matos, T., and Reis, J. 2009. Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems*. 47(4):547–553.
- [4] Cortez, P., Cerdeira, A., Almeida, F., Matos, T., and Reis, J. 2009. UCI Machine Learning Repository Wine Quality Data. URL: <http://archive.ics.uci.edu/ml/datasets/Wine+Quality>.
- [5] Das, S., Rahman, A., Ahamed, A., and Rahman, S. T. 2018. Multi-level models can benefit from minimizing higher-order variations: an illustration using child malnutrition data. *Journal of Statistical Computation and Simulation*, 1-21. <https://doi.org/10.1080/00949655.2018.1553242>.
- [6] Friedman, J., Hastie, T., and Tibshirani, R. 2010. Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*. 33: 1–22.
- [7] Zhang, H., Wahba, G., Lin, Y., Voelker, M., Ferris, M., Klein, R., and Klein, B. 2002. Variable selection and model building via likelihood basis pursuit. Technical Report 1059, University of Wisconsin, USA.
- [8] Hastie, T., Tibshirani, R., and Friedman, J. 2009. The elements of statistical learning: Data mining, inference, and prediction. Springer, NY.
- [9] Hastie, T., Tibshirani, R., and Friedman, J. 2001. The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Springer-Verlag, NY.
- [10] Hastie, T., Tibshirani, R., and Wainwright, M. 2015. Statistical learning with sparsity: The Lasso and generalizations. Chapman Hall, London.
- [11] Guyon, I., and Elisseeff, A. 2003. An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3:1157–1182.
- [12] James, J., Witten, D., Hastie, T., and Tibshirani, R. 2013. An Introduction to Statistical Learning with Applications in R. Springer, NY.
- [13] Legin, A., Rudnitskaya, A., Luvova, L., Vlasov, Y., Natale, C., and D'Amico, A. 2003. Evaluation of Italian wine by the electronic tongue: recognition, quantitative analysis and correlation with human sensory perception. *Analytica Chimica Acta*, 1: 33–34.
- [14] Liu, H. and Zhang, J. 2009. Estimation consistency of the group Lasso and its applications. *J Mach Learn Res Workshop Conf Proc*. 5:376–83.

- [15] Mayoaran, T. 2018. Gradient-based optimization algorithm for ridge regression by using R. *International Journal of Research and Scientific Innovation*. 5(4): 38-44.
- [16] Meier, L., Van de Geer, S., and Bühlmann, P. 2008. The group Lasso for logistic regression. *J R Stat Soc Ser B*. 70:53–71.
- [17] Meinshausen, N. and Yu, B. 2009. Lasso-type recovery of sparse representations for high-dimensional data. *Ann Stat*. 37(1):246–270.
- [18] Montgomery, D. and Peck, V. 2012. Introduction to linear regression analysis (5th ed). John Wiley Sons, Inc., London.
- [19] Rahman, A. 2008. Bayesian predictive inference for some linear models under Student-t errors. VDM-Verlag, Saarbrücken.
- [20] Rahman, A. 2017. *Microdata versus Big Data: Computational and Inferential Challenges in Data Science*. 1. Paper presented at 7th International Conference on Cloud Computing, Data Science and Engineering, Amity University, Noida, India.
- [21] Rahman, A. and Harding, A. 2016. Small area estimation and microsimulation modelling. CRC Press, Florida.
- [22] Rahman, A., Nimmy, S. F., and Sarowar, G. 2018. Developing an automated machine learning approach to test discontinuity in DNA for detecting tuberculosis. In *Proceedings of the Twelfth International Conference on Management Science and Engineering Management* (pp. 277-286). Springer. https://doi.org/10.1007/978-3-319-93351-1_23
- [23] Sardy, S., Bruce, S., and Tseng, P. 1998. Block coordinate relaxation methods for nonparametric signal denouncing with wavelet dictionaries. *Technical report*, Seattle, WA.
- [24] Turban, E., Sharda, R., Aronson, J., and King, D. 2007. Business intelligence, a managerial approach. Prentice-Hall, NY.

8. APPENDIX

Table A1: Coefficient estimates for the white wine data for OLS regression techniques-test set.

	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_4$
OLS	195.2000	0.1090	-1.9640	-0.0313	0.1003
Ridge $_{\lambda_{best}}$	58.6598	-0.0163	-1.6787	0.0781	0.0383
Ridge $_{\lambda_{lse}}$	35.1504	-0.0335	-1.3357	0.1013	0.0184
Lasso $_{\lambda_{best}}$	111.0909	0.0253	-1.7598	0.0730	0.0632
Lasso $_{\lambda_{lse}}$	2.8721	-0.0371	-1.5942	0.0000	0.0091
	$\hat{\beta}_5$	$\hat{\beta}_6$	$\hat{\beta}_7$	$\hat{\beta}_8$	$\hat{\beta}_9$
OLS	0.0531	0.0037	0.0000	-195.8	0.7683
Ridge $_{\lambda_{best}}$	-1.1808	0.0041	-0.0008	-57.0477	0.3896
Ridge $_{\lambda_{lse}}$	-2.0395	0.0035	-0.0009	-32.2148	0.2557
Lasso $_{\lambda_{best}}$	-0.5282	0.0037	-0.0005	-0.0112	0.6019
Lasso $_{\lambda_{lse}}$	-0.4751	0.0020	0.0000	0.0000	0.0527
	$\hat{\beta}_{10}$	$\hat{\beta}_{11}$			
OLS	0.5832	0.1555			
Ridge $_{\lambda_{best}}$	0.6090	0.2562			
Ridge $_{\lambda_{lse}}$	0.4986	0.2110			
Lasso $_{\lambda_{best}}$	0.6957	0.2214			
Lasso $_{\lambda_{lse}}$	0.3258	0.3091			